Stage 2: Completing Implementation & Research

After successfully completing **Stage 1**, which included frontend development and basic ARCore-based floor detection with manual map creation, we moved to **Stage 2**. The objective of this stage was to complete the core implementation of indoor navigation, integrate sensor-based positioning, conduct a detailed research survey, and prepare scholarly documentation for conference submission. This stage focused equally on **technical implementation** and **research validation**.

Week 1: Firebase Authentication Setup and Security Implementation

Overview:

During this week, the primary focus was on implementing a **secure and reliable Firebase Authentication system** for the Android application. Authentication was a critical requirement to ensure that only authorized users could access important features such as station selection, indoor maps, and AR-based navigation. The objective was to establish a strong security foundation while maintaining a smooth and user-friendly login experience.

Progress:

- Integrated **Firebase SDK** into the Android project using Android Studio.
- Enabled **Email and Password Authentication** in the Firebase Console.
- Implemented the **user signup module**, allowing new users to register using valid email IDs and passwords.
- Added password strength validation to prevent weak or insecure credentials.
- Stored essential user profile information securely in **Firebase Firestore** after successful registration.
- Developed the **login functionality** with proper validation for incorrect email formats, wrong passwords, and unregistered users.
- Implemented the **Forgot Password** feature, enabling users to reset their passwords through an email-based verification link provided by Firebase.
- Conducted extensive testing of authentication workflows, including successful registration, login failure scenarios, password reset flow, and session persistence.

Summary:

By the end of this week, Firebase Authentication was fully implemented and thoroughly tested. The application now supports secure user registration, login, and password recovery mechanisms. Access to core application features was restricted to authenticated users only, significantly improving the overall security and reliability of the system.

Next Week and Beyond:

In the upcoming week, the focus will shift toward implementing **ARCore-based navigation features**, including manual indoor mapping and checkpoint-based AR guidance. Initial testing of ARCore environment scanning will also be carried out.

Week 2: Manual ARCore Navigation with Checkpoints and Arrows

Overview:

In this week, the focus shifted towards implementing **manual ARCore-based indoor navigation** using checkpoints and directional arrows. Since GPS signals are unreliable in indoor environments such as railway stations, a predefined and controlled navigation approach was adopted. The objective was to guide users accurately through indoor paths using AR visuals without relying on external positioning infrastructure.

Progress:

- Designed a **manual indoor map** defining source, destination, and multiple intermediate checkpoints.
- Identified checkpoints as key turning points and decision locations along the navigation path.
- Integrated **ARCore anchors** at each checkpoint to establish fixed reference points in the environment.
- Implemented **AR directional arrows** to visually guide users from one checkpoint to the next.
- Addressed arrow instability and flickering issues by applying **pose smoothing techniques** and stable anchor tracking using ARCore's motion tracking.
- Programmed arrows to appear at specific distances and orientations based on straight paths and turning directions.
- Developed a **developer mode** that allows administrators to scan the environment, place anchors manually, and save Points of Interest (POIs).
- Conducted indoor testing to validate arrow visibility, alignment, and navigation flow.

Summary:

By the end of Week 2, a fully functional manual AR navigation system was successfully implemented. Users were able to select a source and destination and follow AR arrows through predefined checkpoints, ensuring reliable indoor navigation without GPS dependency.

Next Week and Beyond:

In the following week, the focus will move towards conducting a **research survey on existing indoor navigation techniques** and analyzing different positioning methods to justify the selected ARCore-based approach.

Week 3: Research Survey on Indoor Navigation Technologies

Overview:

Week 3 was fully dedicated to conducting an **in-depth research and literature survey** on existing indoor navigation and positioning technologies. The main objective was to analyze current solutions, understand their strengths and limitations, and scientifically justify the selection of the proposed approach. This research phase was essential to support the technical decisions with validated academic and industry references.

Progress:

- Studied and analyzed various indoor navigation techniques, including **Wi-Fi fingerprinting**, **Bluetooth Low Energy (BLE) beacons**, and **Inertial Measurement Unit (IMU)-based navigation**.
- Evaluated Wi-Fi-based positioning systems, noting their average accuracy of **4–8 meters** and the requirement of extensive infrastructure setup and maintenance.
- Analyzed BLE beacon-based systems, which provide slightly improved accuracy (**3–5 meters**) but involve additional costs for beacon deployment and calibration.
- Reviewed more than **10 research papers** related to AR-based indoor navigation, sensor fusion methods, and pedestrian dead reckoning techniques.
- Studied **step detection algorithms** using accelerometer data and **heading estimation** using gyroscope sensors.
- Observed that IMU-based navigation achieved **93–97% step detection accuracy** without requiring external hardware or infrastructure.
- Documented comparative analysis using accuracy tables, advantages, limitations, and feasibility metrics for different techniques.

Summary:

The literature survey clearly demonstrated that **ARCore combined with IMU-based step counting** is the most cost-effective, scalable, and infrastructure-free solution for indoor navigation. The findings strongly supported the proposed system design and provided a solid research foundation for implementation and documentation.

Next Week and Beyond:

In the upcoming week, the focus will shift towards implementing **manual step counting and direction estimation** using accelerometer and gyroscope sensors based on the insights gained from the research survey.

Week 4: Manual Step Counting and Direction Estimation

Overview:

In Week 4, the focus was on implementing **manual step counting and direction estimation** using smartphone sensors to enable accurate indoor positioning. Since GPS is ineffective indoors, this module was designed to estimate user movement using sensor data, forming a crucial part of the indoor navigation system. The goal was to achieve reliable distance and direction tracking while minimizing positional drift.

Progress:

- Implemented **step detection** using the smartphone accelerometer by applying peak detection on the magnitude of acceleration signals.
- Utilized the **gyroscope sensor** to detect heading direction and turning movements during navigation.
- Applied a **stride length estimation model** to convert step count into approximate distance traveled.
- Integrated **speedometer data** to dynamically refine distance per step and improve overall accuracy.
- Introduced **manual checkpoints** to periodically reset position estimation and reduce cumulative error.
- Ensured continuous tracking of step count and heading direction between checkpoints.
- Updated **AR directional arrows dynamically** based on the estimated user position and movement.
- Conducted extensive testing with different phone placements such as handheld, pocket, and bag.
- Achieved over **95% step detection accuracy**, validating the robustness and reliability of the approach.

Summary:

By the end of Week 4, a reliable sensor-based navigation module was successfully implemented. The combination of accelerometer, gyroscope, and speedometer data enabled accurate step counting and direction estimation, forming the backbone of the indoor navigation system.

Next Week and Beyond:

In the following week, the focus will be on **full sensor integration and end-to-end testing**, combining step counting, direction estimation, and ARCore navigation into a unified system.

Week 5: Full Sensor Fusion and End-to-End Testing

Overview:

In Week 5, the focus was on **integrating all sensor modules into a single, unified navigation pipeline**. This phase aimed to ensure smooth coordination between sensor-based positioning and AR-based visual guidance. The objective was to validate the complete indoor navigation workflow under realistic conditions and evaluate overall system accuracy.

Progress:

- Integrated **accelerometer-based step counting**, **gyroscope-based heading detection**, and **speedometer-based distance estimation** into a single navigation framework.
- Combined sensor data with **ARCore visual guidance** to provide real-time navigation feedback.
- Defined **manual navigation paths** using a sequence of checkpoints, each linked with predefined step counts and turn directions.
- Implemented navigation instructions such as *"Walk straight for 40 steps"* and *"Turn right after 15 steps"*, supported by AR directional arrows.
- Conducted **end-to-end testing** on a 100-meter indoor path simulating a real railway station environment.
- Used checkpoint-based position resets along with ARCore tracking to minimize cumulative drift.
- Measured system accuracy and observed a **positional error of only 2–3 meters**, outperforming Wi-Fi and BLE-based indoor navigation approaches.

Summary:

By the end of Week 5, the complete sensor fusion-based navigation system was successfully validated. The integration of multiple sensors with ARCore resulted in accurate, reliable, and infrastructure-free indoor navigation, confirming the technical feasibility of the proposed solution.

Next Week and Beyond:

In the upcoming week, efforts will focus on **application-level integration**, user interface refinements, and preparation of demonstration materials for final evaluation and presentation.

Week 6: Application Integration and Demo Preparation

Overview:

During Week 6, the primary focus was on **integrating all developed modules into a complete and user-friendly Android application**. This phase aimed to ensure smooth interaction between authentication, navigation logic, sensor data processing, and AR visualization. The objective was to finalize the application workflow and prepare demonstration materials for evaluation and presentation.

Progress:

- Integrated all system components into a single application flow: **user login → station selection → route selection → AR navigation**.
- Enhanced the **user interface** to clearly display step count, remaining distance, and navigation alerts.
- Implemented a **recalibration feature** at checkpoints using ARCore anchors to improve alignment and navigation accuracy.
- Performed **multi-device testing** across different Android versions and hardware configurations to ensure application stability.
- Identified and fixed minor UI and navigation inconsistencies during testing.
- Recorded a **complete demo video** showcasing the navigation process from source to destination using AR guidance.
- Documented technical learnings and performed a comparative evaluation of different indoor navigation technologies, including Wi-Fi, BLE, and IMU-based approaches.

Summary:

By the end of Week 6, the application was fully integrated, stable, and user-ready. All major functionalities were working seamlessly, and demonstration assets were successfully prepared for final evaluation and academic documentation.

Next Week and Beyond:

In the final week, the focus will shift toward **research paper writing, PPT preparation, and conference submission**, along with final reviews of documentation and results.

Week 7: Research Paper Writing, PPT Creation, and Conference Submission

Overview:

In the final week, the focus was on **documentation, result consolidation, and academic dissemination** of the completed work. The objective was to formally present the proposed indoor navigation system through a research paper and presentation, highlighting both technical implementation and validated research outcomes.

Progress:

- Prepared a complete research paper titled **“Indoor Navigation System Using Sensors: A Low-Computation Hybrid Approach for Railway Stations”**
- Documented the **problem definition**, highlighting challenges of indoor navigation in large railway stations.
- Included a detailed **literature survey** comparing Wi-Fi, BLE, and IMU-based navigation approaches.
- Described the **system architecture**, module interactions, and application workflow.
- Explained implementation details covering ARCore integration, sensor fusion, step counting, and checkpoint-based navigation.
- Presented **experimental results**, including step detection accuracy, positional error analysis, and real-time performance.
- Added performance comparisons demonstrating the advantages of the proposed system over Wi-Fi and BLE-based methods.
- Created a professional **PowerPoint presentation** featuring architecture diagrams, flowcharts, screenshots, and result tables.
- Submitted the research paper and presentation to a **student research conference** for evaluation and presentation.

Summary:

Week 7 successfully concluded Stage 2 of the project with comprehensive documentation and academic submission. The project resulted in a fully functional AR-based indoor navigation application supported by validated research findings.